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A domain-enhanced multi-agent framework for trustworthy literature screening in water research

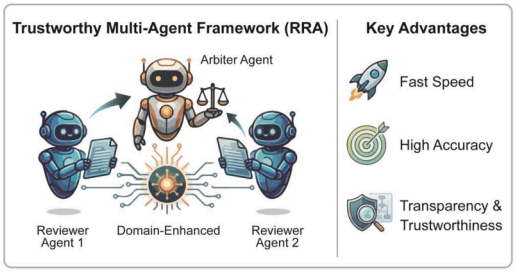
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HIGHLIGHTS

- Developed a domain-enhanced 3T+RAG strategy for LLM-based literature screening.
- Achieved an average F1-score of 0.88 across three water-treatment domains.
- Introduced a Reviewer-Reviewer-Arbiter agent framework for transparent decisions.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Sustainable water management
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ABSTRACT

The integration of artificial intelligence (AI) with sustainable water management holds transformative potential for addressing global sustainability challenges. However, this progress is critically hindered by the slow, labor-intensive construction of large-scale datasets, particularly in identifying the relevant literature from thousands of candidates. Here, we present a hierarchical AI framework that ensures both high accuracy and transparency in literature screening. First, we develop a domain-tailored prompting strategy (3T+RAG) that grounds large language models (LLMs) in structured water-treatment knowledge, thereby improving classification accuracy and screening reliability. Evaluated on three manually curated domain datasets, LLMs achieved reliable literature screening with an average F1-score of 0.88, while operating 79 times faster at only 1.4% of the estimated cost of individual human annotation. To further enhance transparency and auditability, we introduce a multi-agent Reviewer-Reviewer-Arbiter (RRA) framework, in which two reviewer agents independently assess literature using the 3T+RAG prompting strategy, and an arbiter agent resolves disputes through reasoning-trace analysis. This architecture distinguishes reviewer disagreements from consistent decisions, allowing human review to focus on disputed articles and targeted quality control. Overall, this study establishes an efficient and reliable pathway for water-treatment literature screening, providing a robust foundation for large-scale dataset construction and accelerating sustainable water management.

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1. Introduction

Artificial intelligence (AI) is increasingly reshaping sustainable water management by enabling intelligent monitoring, predictive modeling, and data-driven decision-making across complex water systems (Huang et al., 2021; Lu et al., 2025). By learning complex, non-linear relationships from high-dimensional scientific and operational data, AI can generate predictions and insights that are often unattainable through conventional approaches (Cui et al., 2025; Montfort et al., 2025; Wang et al., 2023). Recent advances in machine learning have accelerated progress in water purification, including membrane processes, electrochemical treatment, and adsorption technology (Jeong et al., 2024; Sun et al., 2023; Xenitopoulos et al., 2026; L. Yuan et al., 2025), demonstrating the transformative potential of AI-assisted scientific discovery in water research. These developments offer new opportunities for improving water quality, resource efficiency, and operational sustainability, thereby contributing to global water security and climate resilience.

However, the full potential of AI in water research is limited by the slow construction of large-scale datasets (Fiil-Flynn et al., 2022; Zhu et al., 2023). Although decades of water-treatment studies have produced vast amounts of experimental knowledge, much of this information remains buried within unstructured scientific literature (e.g., pollutant concentrations (Wu et al., 2026) and removal efficiency (Sun et al., 2023)). To support AI-assisted modeling and intelligent decision-making, these scattered data must first be systematically identified, screened, and transformed into structured datasets (Schilling-Wilhelmi et al., 2025; Wu et al., 2026). The conventional workflow for this task, however, is labor-intensive and time-consuming, often taking months or even years, which critically hinders AI-driven discovery (Jiang et al., 2024; Suvarna et al., 2023). An important challenge in dataset construction arises during the initial identification of relevant articles. Conventional database searches (e.g., Web of Science) are essential for broad and high-recall retrieval, but their reliance on keyword and Boolean matching limits their ability to resolve complex semantic relationships, resulting in many irrelevant candidates and low precision (Delgado-Chaves et al., 2025). Consequently, substantial manual effort is often required to identify studies that satisfy predefined inclusion criteria. In addition, the exponential growth of scientific articles (Fig. S1) further exacerbates this challenge, rendering exhaustive manual screening impracticable (Delgado-Chaves et al., 2025; Montfort et al., 2025). Therefore, developing reliable and scalable AI-assisted literature-screening systems is increasingly important for intelligent water research and sustainable water management.

Conventional natural language processing (NLP) and keyword-based approaches have been widely used for literature retrieval and information extraction in scientific research (Wang et al., 2019). While these methods are effective for identifying explicit patterns and broad thematic relevance, they often struggle with context-dependent terminology, ambiguous technology boundaries, and implicit experimental relevance across water-treatment studies (Cheng and Zhang, 2025). Large language models (LLMs), a transformative natural language processing technology, provide new opportunities for understanding and utilizing information in scientific literature (Schilling-Wilhelmi et al., 2025; Pruyn et al., 2025; Suvarna and Pérez-Ramírez, 2024). Their success in material discovery (Yuan et al., 2025; Zeni et al., 2025), medical diagnosis (McDuff et al., 2025; Pais et al., 2024), and other fields (Luo et al., 2025; Stoppa et al., 2025) confirms their ability to handle complex scientific texts. Current applications of LLMs in the environmental field focus on mining data from the literature on specific topics (e.g., chemical toxicology (Liang et al., 2024), and single-atom catalysis (Wu et al., 2026)). However, such studies have typically skipped the time-consuming article screening step and relied on manually prescreened subsets of articles. To date, the ability of LLMs to perform reliable, high-throughput screening of water-related environmental articles remains largely underexplored. Existing AI-assisted screening studies (Cheng and Zhang, 2025; Chibwe et al., 2025) have demonstrated the feasibility of using LLMs for article screening, but two limitations remain. First, screening is generally formulated as a single binary relevance decision, making it difficult to determine which eligibility criterion leads to inclusion or exclusion. In addition, these approaches have not been systematically benchmarked across diverse water-treatment themes. Second, most existing systems rely on a single model and therefore lack an explicit mechanism for detecting disagreements or enabling targeted review before downstream analysis.

Herein, we propose a trustworthy multi-agent framework for automated literature screening in water research, which integrates the structured prompt strategy, retrieval-augmented generation (RAG), comprehensive benchmarking, and collaborative AI reasoning. First, we developed a domain-enhanced 3T+RAG prompting strategy that combines structured screening criteria with water-treatment knowledge to improve classification reliability. We then evaluated the resulting strategy across three representative water-related benchmark datasets (i.e., membrane separation, electrochemical oxidation, and adsorption) and benchmarked its performance against human annotators in terms of accuracy, speed, and cost. To further enhance trustworthiness, we developed a multi-agent Reviewer-Reviewer-Arbiter (RRA) framework, which detects traceable errors through dual-reviewer evaluation and resolves them using an arbiter agent (Fig. 1). Overall, this work establishes a practical pathway for AI-assisted knowledge management and intelligent scientific workflows in sustainable water research.

2. Methods

2.1. Dataset collection and ground-truth construction

The benchmark datasets were constructed through a systematic multi-stage process designed to ensure diversity, quality, and representativeness across three key water treatment themes: membrane separation for PFAS removal, biochar adsorption for heavy metal removal, and electro-oxidation for organic pollutant degradation. The construction workflow encompassed systematic article retrieval from the Web of Science Core Collection using theme-specific advanced search strategies (Text S1), followed by rigorous data extraction and cleaning protocols, in which records missing critical metadata (e.g., abstract) were excluded to maintain data integrity. All retained articles underwent manual annotation based on the 3T (Target-Technology-Type) criteria, and the resulting labels served as the ground truth for benchmarking. To guarantee annotation reliability, a rigorous quality assurance protocol was implemented through multi-stage independent review and consensus procedures. This process yielded three high-quality curated datasets—Dataset I (total N=371), Dataset II (total N=2,958), and Dataset III (total N=5,852)—with comprehensive metadata including titles, abstracts, 3T labels, and DOIs to support reproducible research. For tractable evaluation, stratified sampling was employed to create representative subsets (n=330) from the larger datasets (II and III) while maintaining consistent positive-to-negative ratios, ensuring robust and comparable performance assessment. Dataset I (n=371), sampled Dataset II (n=330), and sampled Dataset III (n=330) were used for all benchmarking in this study.

To establish a reliable ground-truth dataset for benchmarking, a multi-stage expert review protocol was implemented to account for the complexity and ambiguity of literature screening in water research. Each article was independently evaluated by three experts. Conflicting labels were resolved through structured discussion, and the resulting labels were used as the ground truth. This protocol was designed to address task ambiguity and ensure that benchmark labels reflect collective expert judgement rather than individual opinion.

2.2. Large language models (LLMs)

We implemented both the single-LLM and multi-agent frameworks in Python using LLM application programming interfaces (APIs). The models used in this study were ChatGPT-4.1, Claude-3.5-haiku, Grok-3-mini, DeepSeek-V3, and Moonshot-v1-8k. The temperature parameter was set

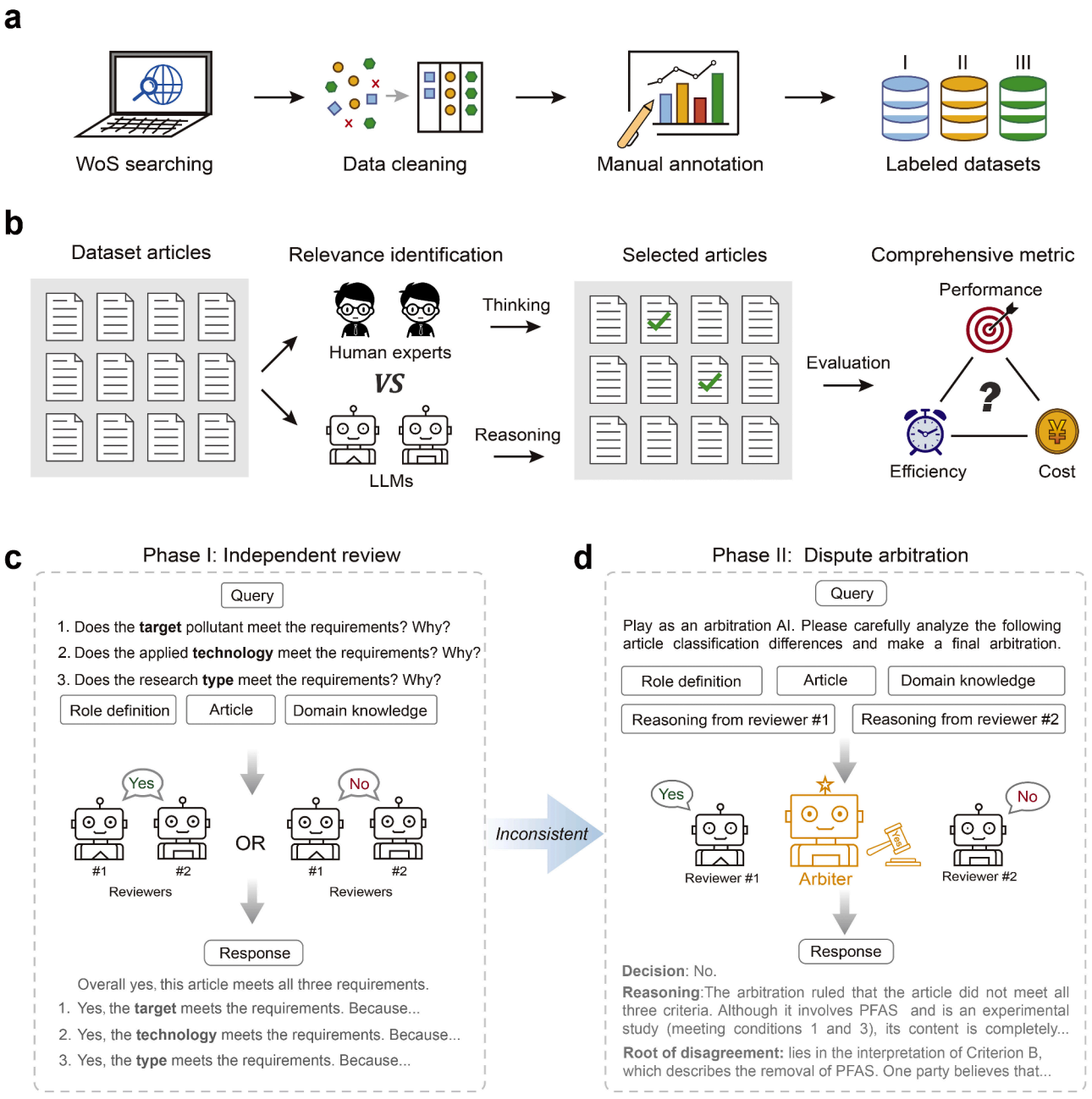


Fig. 1. Multi-agent collaborative framework for article screening and evaluation. **a** Workflow for dataset construction and annotation. The process begins with systematic retrieval from the Web of Science (WoS) database, followed by rigorous data cleaning and manual annotation by human annotators to generate a high-quality, labeled dataset. **b** Comparative theme identification and selection module. A subset of articles is evaluated in parallel by human annotators and five LLMs (ChatGPT, Claude, Grok, DeepSeek, and Moonshot). Their performance in identifying relevant themes is compared and quantitatively assessed using a comprehensive set of metrics encompassing performance, efficiency, and cost. **c** Phase I: independent dual-review process. Two LLM-based reviewers, operating under defined roles and domain knowledge, independently assess each article against three predefined criteria. Each assessment includes a binary decision ("Yes"/"No") and supporting reasoning. Consistent decisions are directly accepted as the finial decision. **d** Phase II: arbitration mechanism for resolving discrepancies. When the primary reviewers reach inconsistent conclusions, a dedicated arbitration AI module is activated. This module analyzes the original articles, the conflicting decisions, and the accompanying reasoning from both reviewers to deliver a final judgement, thereby ensuring the reliability of the screening outcome.

to zero for all models to minimize the randomness of model outputs. Detailed programming environments and package versions used are provided in Text S2. The input to each LLM comprised a system prompt and a user prompt. The system prompt defined the role of each agent (e.g., reviewer or arbiter) for water research and environmental remediation tasks. The user prompt provided a specific description of tasks, including screening criteria, format requirements, article information, and background knowledge. The LLMs were required to structure their responses in two parts: judgement (restricted to 'Yes' or 'No') and reasoning (which required a logical process supported by evidence). This design facilitates the programmatic parsing, extraction, and organization of key information from the output. More information about LLMs is provided in Text S3. Reviewer-agent outputs were further subjected to word-frequency analysis after custom stopword filtering (Text S4).

2.3. Structured 3T prompt strategy and retrieval-augmented generation (RAG)

According to the characteristics of data-driven water research, we designed a structured 3T prompt strategy tailored to water-treatment and environmental-remediation research. Rather than asking the model to provide a single overall relevance judgement, the prompt decomposes article screening into three separate decisions: (T1) research target alignment (e.g., heavy metal removal from water); (T2) technology match (e.g., biochar adsorption); and (T3) study type qualification (e.g., experimental research). An article is included only when all three criteria are satisfied. This design requires separate assessments of Target, Technology, and Type and records the basis for each judgement, allowing classification errors to be traced to specific screening criteria.

To enhance the domain grounding, we implemented a retrieval-augmented generation (RAG) module using a task-specific water-treatment knowledge repository (Text S5). The repository was manually curated from review articles and domain reference materials related to membrane separation, electrochemical oxidation, and adsorption processes. Curation was conducted independently of the benchmark articles, ground-truth labels, and model predictions. It was not intended as a large-scale corpus; instead, it contained concise entries describing domain definitions, technology boundaries, and element classifications relevant to the benchmark tasks. The repository contained approximately 50 curated entries and was constructed before model benchmarking. Each curated knowledge entry was treated as an independent retrieval unit and encoded separately using the *distiluse-base-multilingual-cased-v1* sentence-transformer model. Query embeddings were generated from article titles and abstracts, and semantic retrieval was performed using cosine similarity. The most relevant entries were appended to the prompt as contextual domain knowledge to support article classification. Since the repository primarily consisted of concise task-specific definitions and terminology, only limited manual curation was required prior to benchmarking. The repository was intentionally designed as a compact, task-specific resource rather than a comprehensive knowledge base. When the framework is applied to a new research topic, researchers only need to construct a similarly sized repository containing the key definitions and terminology of that field, enabling low-cost transfer without developing a large general-purpose knowledge base.

2.4. Human and LLM performance evaluation

Human performance was evaluated using a cohort of nine annotators, including three postdoctoral researchers, three doctoral researchers, and three master's researchers. All annotators screened the articles independently under standardized conditions, and their individual decisions were evaluated against the ground-truth labels. To assess the effect of majority voting, the annotators were further divided into three experience-balanced groups. For each article, a group-level decision was obtained by majority voting among the three independent annotators. Individual human annotators, three-annotator human groups, and individual LLMs were evaluated against the ground truth for a fair comparison.

We evaluated the accuracy of humans and LLMs on the three sampled datasets using three standard metrics: precision, recall, and F1-score. The sampling methodology and the detailed testing protocol are described in Texts S6 and S7, respectively. The metrics are calculated as follows (Turkina et al., 2025):

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$\text{F1-score} = 2 \times \frac{Precision \cdot Recall}{Precision + Recall}$$

where TP (True Positives) is the number of instances correctly identified as relevant, and FP (False Positives) is the number of instances incorrectly identified as relevant. FN (False Negatives) denotes the number of instances that are relevant but are incorrectly identified as irrelevant.

For time-efficiency evaluation, 20 parallel threads were adopted to balance computational throughput and system stability. Preliminary testing indicated that increasing thread numbers improved processing efficiency but also introduced higher variability in API response latency and occasional request failures.

Human screening cost was estimated from the measured annotation time and the annual compensation of the participating researcher groups. Based on an average screening time of 39.64 s per article and 2,000 working hours per year, the weighted cost of a professional human annotator was approximately 0.65 CNY per article (Text S8). The cost of LLMs was calculated according to the tokens consumed and the official API price, as follows:

$$Cost = Price_{in} \times Tokens_{in} + Price_{out} \times Tokens_{out}$$

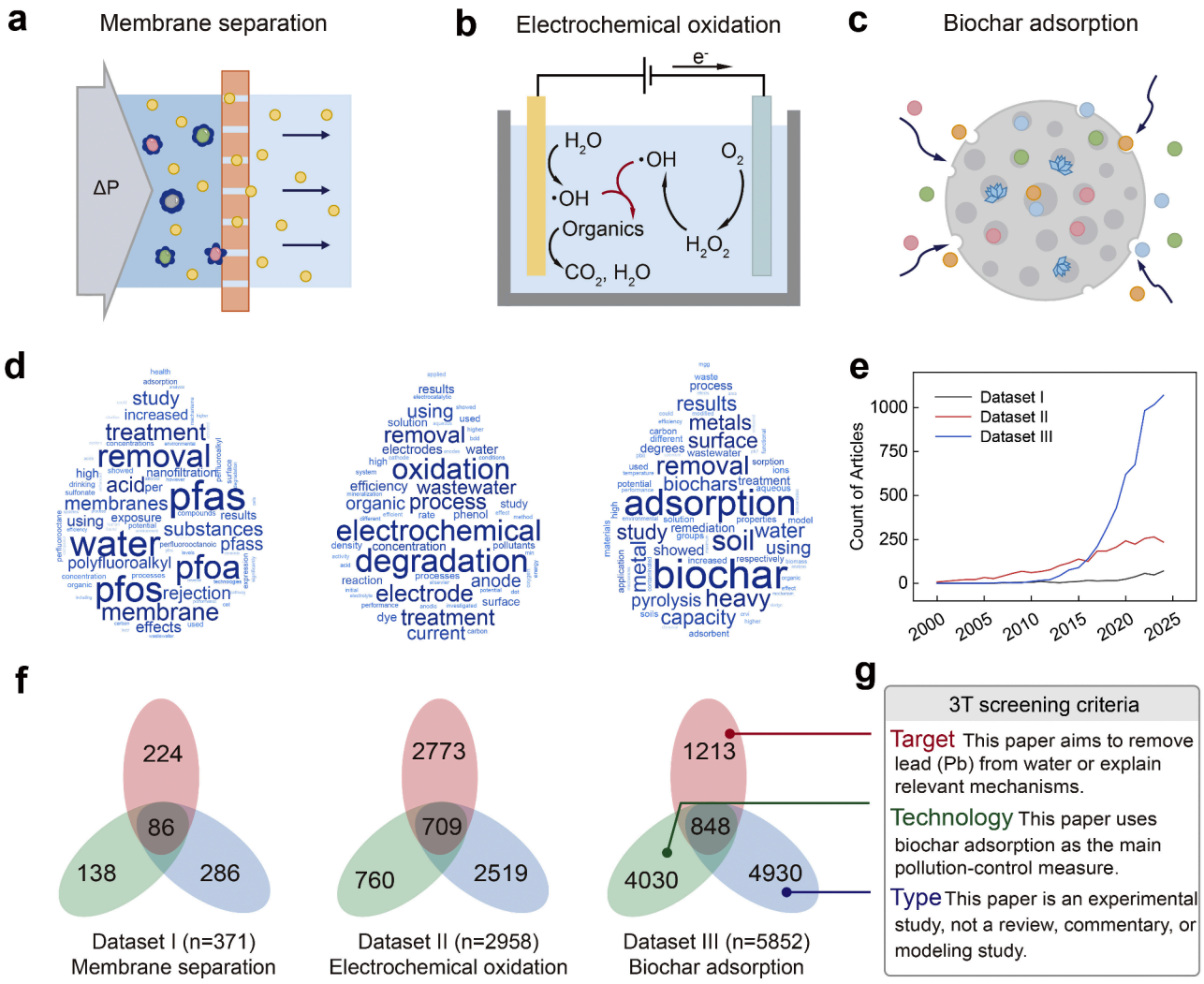
where Price_{in} represents the cost per token for the input to the LLM, $\text{Tokens}_{\text{in}}$ is the number of input tokens processed by the LLM, $\text{Price}_{\text{out}}$ denotes the cost per token for the output generated by the LLMs, and $\text{Tokens}_{\text{out}}$ is the number of output tokens produced by the LLM. The number of tokens was calculated using the Python package *tiktoken* provided by OpenAI. The fixed input components, including the system prompt and task instructions, contained approximately 300 tokens, which is below the commonly used cacheable length thresholds of most LLM providers. Therefore, cache discounts were not considered, and standard input-token pricing was adopted to provide a consistent and conservative cost estimation across different models.

3. Results and discussion

3.1. Article dataset characteristics

To evaluate the recognition abilities of LLMs and humans, we constructed three benchmark datasets across distinct water-treatment and environmental-remediation domains. These datasets comprise hundreds to thousands of scientific articles that were obtained through extensive keyword-based searches. Subsequently, all articles were manually reviewed and annotated as relevant or irrelevant to establish ground-truth labels for performance evaluation. We also designed typical application scenarios for each water-treatment technology: membrane separation for per- and polyfluoroalkyl substances (PFAS) removal (Fig. 2a), electrochemical oxidation for organic pollutant degradation (Fig. 2b), and biochar adsorption for heavy metal removal (Fig. 2c). These datasets reflect real-world research scenarios and were validated to ensure thematic consistency and relevance. Statistical analysis of high-frequency vocabulary demonstrated strong thematic clustering, with each dataset focused on its respective core concepts (Fig. 2d and Table S1). Temporal analysis showed a significant increase in publication volume in recent years, highlighting the timeliness of these research areas and the need for efficient screening tools (Fig. 2e).

A critical challenge in dataset construction was the extensive refinement process from initial search results to final qualified entries (Delgado-Chaves et al., 2025). When using LLMs to screen literature, the quality of the prompting strategy directly affects accuracy, making it necessary to standardize and optimize prompt design (Zheng et al., 2024, 2023). Based on the characteristics of water-treatment research, we proposed a structured 3T (Target-Technology-Type) prompt strategy, which evaluated articles across three dimensions: (1) target pollutant and medium, which specify the pollutant classes (e.g., antibiotics, PFAS) and environmental compartments (e.g., water, soil) of interest; (2) technology applied, which delineates the remediation techniques of interest (e.g., individual or coupled processes) while excluding similar but irrelevant methods to maintain technical consistency; and (3) research type, which



h

No.	Journal	Title	Abstract	Label	LA	LB	LC	DOI
1	*Chem. Eng. J.*	Synergistic adsorption-deg...	Compared to traditional single pollu...	0	1	0	1	10.1016/j.cej.2024...
2	*Chemosphere*	Enhanced cadmium immo...	Cd toxicity emerges as a major envi...	0	0	1	1	10.1016/j.chemosphere...
3	*Environ. Res.*	Mn oxide-modified biochar...	The occurrence of excessive levels...	1	1	1	1	10.1016/j.envres.2024...
...								
5851	*J. Clean. Prod.*	Interpreting machine learn...	Employing machine learning to pred...	0	1	1	0	10.1016/j.jclepro.2024...
5852	*Bioresour. Technol.*	Emerging applications of bio...	Biomass fast pyrolysis produces bio...	0	0	1	1	10.1016/j.biortech.2023...

Fig. 2. Characteristics and validation of three water-related environmental datasets. a-c Schematic diagrams of three research themes: membrane separation (Dataset I), electrochemical oxidation (Dataset II), and biochar adsorption (Dataset III). In panel **b**, the black arrows denote substance conversion, while the red arrows represent reactions between •OH and organics. **d** Word cloud visualizations of the high-frequency words used in the three datasets. Each panel corresponds to a distinct dataset, with word frequency represented by font size and color. Darker and larger fonts indicate higher frequency. **e** Annual number of published articles for Datasets I, II, and III. **f** Number of articles meeting the 3T criteria: Target (red), Technology (green), and Type (blue). For example, Dataset III (biochar adsorption) includes a total of 5,852 articles, with 1,213 articles meeting the target criteria (red), 4,030 articles meeting the technology criteria (green), 4,930 articles meeting the type criteria (blue), and 848 articles fulfilling all three criteria. **g** Example of the 3T criteria for Dataset III. **h** Structural schematic of Dataset III, including basic article information and annotations. L1, L2, and L3 correspond to the labels for the T1, T2, and T3 criteria, where a value of "1" indicates "Yes" and "0" indicates "No".

focuses on experimental studies that provide primary data, while typically excluding reviews and modeling articles to ensure the quality of data. As shown in Fig. 2f, although the initial article searches yielded thousands of candidate articles, only approximately 20% met the 3T selection criteria, highlighting the challenge of identifying truly relevant articles amidst broadly related research. The 3T selection criteria provided a rigorous framework for article qualification, with annotation statistics showing the distribution of articles meeting various criteria combinations. The intersection represented the small subset (~20%) that satisfied all three conditions simultaneously—the primary target for identification by both LLMs and human annotators (Figs. 2f and g). Each article in the final datasets was annotated with key attributes—including title, abstract, 3T labels, and

digital object identifiers (DOIs)—to support fully reproducible benchmarking (Fig. 2h). To ensure robust evaluation across datasets of varying sizes, stratified sampling was employed to create representative subsets, maintaining consistent positive-to-negative ratios for reliable performance comparison. More details about dataset construction and annotation are provided in the Methods section.

3.2. Domain-enhanced 3T+RAG strategy improves literature-screening reliability

Applying general-purpose LLMs to specialized domains faces challenges due to their tendency to generate incorrect or misleading answers, known as hallucinations, which primarily result from insufficient reasoning capabilities and domain knowledge (Rafiq et al., 2025; Swanson et al., 2025). In literature screening, however, incorrect classifications may also arise from incomplete article information, ambiguous screening criteria, or unclear topic boundaries. To improve screening accuracy and reliability, we developed a 3T+RAG prompting strategy that combines structured screening criteria with relevant water-treatment knowledge. The 3T+RAG prompting strategy consists of two core components. First, the 3T criteria structured the reasoning of LLMs by requiring them to evaluate each article against three dimensions: (T1) whether the research target aligns with a specified water-treatment outcome (e.g., heavy metal removal); (T2) whether the employed technology matches the target process (e.g., biochar adsorption); and (T3) whether the study type qualifies (e.g., experimental research) (Fig. 2g). Second, the retrieval-augmented generation (RAG) component (Mon-Williams et al., 2025) supplies task-relevant domain context through retrieval from a curated water-treatment knowledge repository. A complete 3T+RAG prompt consists of five parts: role definition, task description (3T), article information to be analyzed, output format requirements, and retrieved domain knowledge (Fig. 3a). This standardized structure provides a starting point for applications in neighboring domains. Researchers can retain the five-part prompt structure and revise only the domain-specific criteria and terminology, rather than developing a new screening prompt from scratch. To validate the effectiveness of our approach, we designed a baseline

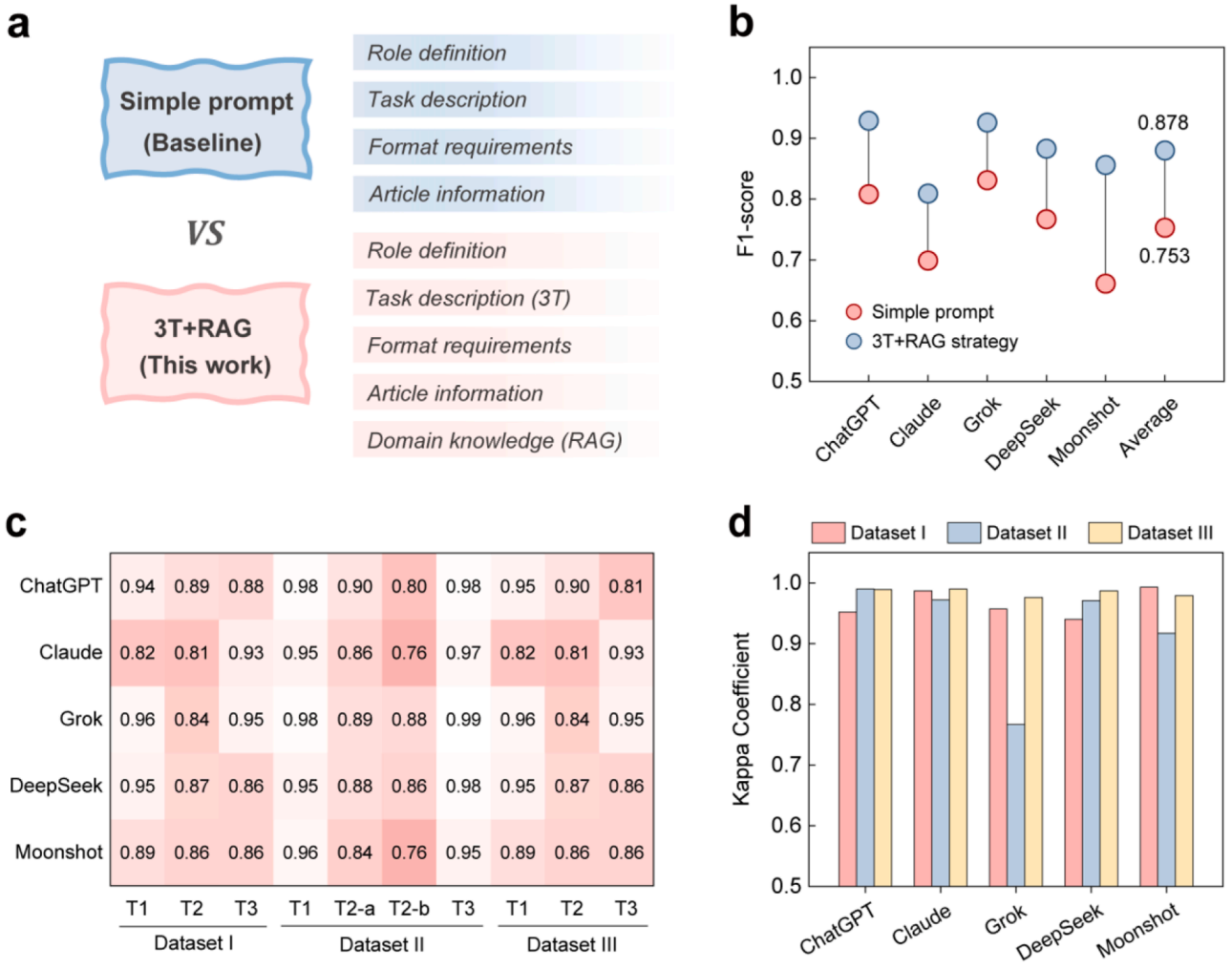


Fig. 3. Performance evaluation of the 3T+RAG prompting strategy across multiple LLMs and datasets. **a** Schematic comparison of the prompt engineering strategies. The baseline method (simple prompt) includes role definition, task description, format requirements, and article information, while the proposed 3T+RAG prompting strategy incorporates structured 3T criteria and contextual information retrieved from a curated water-treatment knowledge repository. **b** Performance improvement measured by F1-score. The 3T+RAG method (blue) consistently outperforms the simple prompt baseline (red) across all evaluated large language models (ChatGPT, Claude, Grok, DeepSeek, and Moonshot). **c** Heatmap depicting the detailed F1-scores for each model across three distinct datasets. T1, T2, and T3 correspond to the three dimensions of the 3T screening criteria: target pollutant, applied technology, and type of research. For Dataset II, the T2 dimension is composed of two sub-criteria: a standalone electro-oxidation system (T2-a) and systems where the anode material contains transition metals (T2-b). **d** Intra-model consistency assessment. Cohen's kappa coefficients measure the agreement in a model's responses to the same questions across two separate trials. A value approaching one indicates high stability in the decision-making process of the model. All models maintain strong consistency under the guidance of the 3T+RAG strategy.

prompt that summarized the task in one sentence and provided no additional domain knowledge (Fig. 3a). We then applied both the 3T+RAG and baseline prompts to five prevalent LLMs: ChatGPT, Grok, Claude, DeepSeek, and Moonshot. The results indicated that the 3T+RAG prompt improved the average F1-score (Itani et al., 2025), a composite metric of classification performance (see Methods for detailed description), across all LLMs by 12.5% (Fig. 3b). This finding shows that explicit screening criteria and relevant domain knowledge improve classification performance across the evaluated models. Moreover, the 3T+RAG prompting strategy facilitated a detailed breakdown of model performance across three dimensions, enabling deeper analysis and further improvement (Fig. 3c). For example, high F1-scores in T1 and T3 demonstrated that the LLMs accurately identified target pollutants and research types. In contrast, T2 involved the recognition of relevant treatment technologies and required more domain knowledge, resulting in comparatively lower scores (Fig. 3c). Additionally,LLMs using the 3T+RAG prompting strategy exhibited high self-consistency across repeated evaluations, with Cohen's kappa coefficients mostly exceeding 0.90 (Fig. 3d), indicating the stability of domain-enhanced LLM-based reasoning.

To illustrate the difference between the 3T+RAG prompt strategy and the baseline prompt, we presented a representative case from Dataset II. This dataset was selected for its stricter thematic constraints, which resulted in the highest difficulty level among the three datasets evaluated. In this case, the LLM was tasked with evaluating whether this study focused on electrochemical oxidation using transition metal anode materials for organic pollutant removal. Under the baseline prompt, the requirement was directly presented to the model, and no additional domain knowledge was supplied (Fig. 4a). In contrast, the 3T+RAG approach decomposed the query into three structured sub-questions (i.e., target, technology, and type) and supplemented them with background knowledge (Fig. 4b). In this case, the retrieved information concerned the periodic-table classification of Sb and Sn. This enabled the model to evaluate the anode material against the predefined technology criterion. The results indicated that the baseline prompt may led to incorrect answers, particularly in identifying transition metals (Figs. 4c and d). By applying the 3T+RAG prompt strategy, the domain-enhanced LLM not only improved the accuracy of article classification but also provided a verifiable reasoning trace (Fig. 4e). Overall, the baseline prompt can result in generic, hallucinated, or incomplete justifications, whereas the 3T+RAG prompting strategy enforces a systematic evaluation against predefined criteria, producing outputs that are more accurate, reliable, and logically grounded. These findings affirm the efficacy of the structured 3T+RAG approach in constraining model reasoning to mitigate hallucinations and enhance evaluative rigor.

To further quantify the contribution of individual components, we performed an ablation analysis comparing a simple prompt baseline, RAG-only prompting, 3T-only prompting, and the complete 3T+RAG strategy (Table S2). Compared with a conventional prompt that requests a single relevance judgement, the structured 3T prompt evaluates Target, Technology, and Type separately, allowing the basis of the final decision to be examined and any misclassification to be traced to a specific screening criterion. The 3T-only strategy substantially improved the average F1-score from 0.753 to 0.849, demonstrating the value of evaluating Target, Technology, and Type separately rather than relying on a single overall relevance judgement. However, the RAG-only strategy yielded a lower average F1-score than the baseline (0.742 vs. 0.753) (Table S2). To further understand the effect of RAG, we analyzed cases where RAG corrected baseline errors and cases where it introduced new errors (Texts S9 and S10). RAG helped most with ambiguous terminology and domain-boundary cases but could hurt when retrieved content was only broadly relevant or distracted from article-specific evidence. This is consistent with the retrieval relevance assessment (~62%). Accordingly, RAG alone did not improve performance, as retrieval without structured reasoning introduced contextual noise. The 3T+RAG strategy achieved the best result (0.878), confirming that domain knowledge is most effective when integrated with structured decision criteria. These findings suggest that the observed performance gains arise from complementary interactions between reasoning structure and contextual knowledge rather than from prompt complexity alone. These comparisons provide a more informative benchmark than single-prompt evaluation and enable stepwise assessment of individual design choices. Notably, the wording and ordering of prompt instructions can influence LLM outputs and, consequently, the accuracy of literature screening. A small manually labelled dataset can therefore be used to refine the criteria until satisfactory performance is achieved before the prompt is applied to the full literature set.

To assess whether the 3T+RAG strategy remains effective as general-purpose models advance, we conducted additional benchmarking using GPT-5.5. The results showed that while simple prompts perform well for studies with unambiguous eligibility boundaries, the 3T+RAG strategy provided substantially higher screening performance when eligibility hinges on strict technical definitions and fine-grained domain distinctions (Table S3). Specifically, in Dataset II, the 3T+RAG strategy corrected 44 errors made by the simple prompt while introducing only three new errors. McNemar's exact test confirmed that this difference was statistically significant ($p < 0.001$), indicating that structured criteria and domain knowledge remained beneficial even with a stronger baseline model.

3.3. Comparative evaluation of screening accuracy, speed, and cost

To validate the comprehensive performance of the domain-enhanced 3T+RAG strategy, we conducted a quantitative analysis of three datasets and benchmarked the models against human annotators. The evaluation cohort included nine human annotators specializing in water and environmental science and the five LLMs described above (ChatGPT-4.1, Grok-3-mini, Claude-3.5-haiku, DeepSeek-V3, Moonshot-v1-8k). Each was evaluated on three key dimensions: accuracy, speed, and economic cost. Across three benchmark datasets, domain-enhanced LLMs powered by the 3T+RAG prompting strategy demonstrated performance comparable to or exceeding that of individual human annotators, while exhibiting substantial advantages in processing speed and cost-effectiveness (Fig. 5a), highlighting their potential to assist large-scale article screening. The human evaluation cohort comprised nine water-treatment researchers spanning multiple levels of research experience. Screening decisions were made using titles and abstracts under standardized conditions and evaluated jointly across Target, Technology, and Type criteria. In addition, Fleiss' kappa indicated moderate agreement among independent annotators (0.52–0.66) and substantial agreement within the ground-truth panel (0.62–0.77), reflecting the inherent ambiguity of the screening task (Tables S4 and S5).

First, we evaluated accuracy using three standard metrics: precision (the ratio of correctly predicted positive articles to the total predicted positive articles (Turkina et al., 2025), reflecting the correctness of positive predictions); recall (the ratio of correctly predicted positive articles to all actual positive articles (Turkina et al., 2025), reflecting comprehensiveness); and F1-score (the harmonic mean of precision and recall (Yao et al., 2025); see Methods for equation). Here, "positive" and "negative" refer to articles that meet or do not meet the screening criteria, respectively. When guided by the domain-enhanced strategy (3T+RAG), LLMs achieved superior precision (0.882, 0.859, and 0.941) compared with individual human annotators (0.763, 0.666, and 0.800) across the three datasets, indicating a stronger capability to accurately identify relevant articles (Fig. 5b). The structured reasoning and retrieval in the 3T+RAG approach also contributed to high recall on Dataset I and Dataset III. Dataset II showed a comparatively lower recall of 0.796, which can be attributed to its stricter criteria and greater proportion of borderline cases (Fig. 5c). This level of recall may be insufficient for recall-sensitive literature screening, particularly for systematic-review-style applications where missed studies are costly. In such cases, the screening criteria should be moderately relaxed to improve recall, with the expected trade-off of lower precision. To provide a more rigorous human benchmark, we further evaluated three-annotator human groups using majority voting. Although consensus improved human screening performance, a single LLM still achieved F1-scores comparable to or higher than those of the consensus-based

a

Simple prompt

You are an expert in literature screening and recommendation in the field of water research and environmental science. Please analyze the provided literature (title and abstract) and context thoroughly and respond strictly according to the following criteria: A laboratory-level experimental study that describes the degradation of organic pollutants in water using transition-metal-based anodic oxidation. Provide a specific response (Yes/No) followed by a justification that cites precise textual evidence. All claims must be verifiable through direct paper and context—avoid speculative answers. Prioritize strict interpretation (e.g., 'possible' = not meeting the criterion). All replies must be in English and follow the required format:

Overall Response: {Yes/No} Reasoning: {Brief justification with citations}

Title: Electrochemical degradation of ciprofloxacin with a Sb-doped SnO2 electrode: performance, influencing factors and degradation ...
Abstract: Sb-doped SnO2 electrodes were prepared with the practical sol-gel method and were used for the electrocatalytic degradation...

b

3T+RAG prompt

You are an expert in literature screening and recommendation in the field of environmental science. Please analyze the provided literature (title and abstract) thoroughly and respond strictly according to the following criteria:

(A) The literature describes or includes the removal of organic pollutants from water OR explains mechanisms for organic pollutant removal from water.
(B1) The used technology is single electrooxidation/electrocatalytic oxidation (not coupled with other technologies such as photocatalysis, PMS activation, ozone catalysis, membrane filtration, etc.).
(B2) This requirement is satisfied if either of the following criteria is met: (i) The substrate of anode contains at least one transition metal element (ii) The active layer of anode contains at least one transition metal element (iii) Other part of anode contains at least one transition metal element.
(C) The literature is a laboratory-level experimental study (excluding reviews, commentaries, perspectives, opinions, modeling articles, pilot-scale or water plant-level studies).

The overall response must be 'Yes' only if above four conditions are all met, else be 'No'. For each criterion, provide a specific response (Yes/No) followed by a justification that cites precise textual evidence. All claims must be verifiable through direct paper and context—avoid speculative answers. All replies must be in English and follow the required format, including line breaks: Overall Response: {Yes/No} Reasoning: {Brief justification with citations} Response A: {Yes/No} Reasoning A: {Cited text} {Explanation} Response B1: {Yes/No} Reasoning: {Explanation} Response B2: {Yes/No} Anode:{Anode materials} Reasoning: {Explanation} Response C: {Yes/No} Reasoning C: {Explanation}.

Title: Electrochemical degradation of ciprofloxacin with a Sb-doped SnO2 electrode: performance, influencing factors and degradation ...
Abstract: Sb-doped SnO2 electrodes were prepared with the practical sol-gel method and were used for the electrocatalytic ...

Domain knowledge (must read):
1. Electrocatalytic membranes (also referred to as electro-filtration or electrified membranes) are not standalone electro-oxidation technologies, but rather coupled processes that integrate membrane filtration with multiple electrochemical phenomena. (Similarity: 0.026)
2. Vanadium (V, atomic number 23) belongs to transition metals. (Similarity: 0.009)
3. Antimony (Sb, atomic number 51) is not a transition metal (Similarity: 0.009)
4. Tin (Sn, atomic number 50) is not a transition metal. Tin dioxide (SnO2) does not belong to transition metal oxides (Similarity: 0.009)

c

DeepSeek (Simple prompt)

Response: Yes
Reasoning: The paper describes a laboratory-level experimental study involving the degradation of ciprofloxacin (an organic pollutant) in water using a transition-metal-based (Sb-doped SnO2) anodic oxidation method. The abstract explicitly mentions the preparation of the electrode, electrochemical characterization, optimization of degradation parameters, and degradation pathways, all of which align with the criteria. Key evidence includes: "Sb-doped SnO2 electrodes were prepared... for the electrocatalytic degradation of ciprofloxacin (CIP) in aqueous solution" and "the removal ratio of CIP reached to almost 100%

d

ChatGPT (Simple prompt)

Overall Response: Yes
Reasoning: The study is laboratory-level, involving experimental preparation and testing ("Sb-doped SnO2 electrodes were prepared...and were used for the electrocatalytic degradation of ciprofloxacin in aqueous solution"). It describes the use of a transition-metal-based electrode (SnO2, doped with Sb—a transition metal) for anodic oxidation of an organic pollutant (ciprofloxacin), with details on performance and degradation pathways ("removal ratio of CIP reached to almost 100% within 60 min...possible degradation pathways of CIP were proposed"). All claims are directly supported by the text.

e

DeepSeek (3T+RAG prompt)

Overall Response: No
Reasoning: The paper does not meet all four conditions (A, B1, B2, C) simultaneously.

Response A: Yes
Reasoning A: The abstract states, "Sb-doped SnO2 electrodes were prepared... and were used for the electrocatalytic degradation of ciprofloxacin (CIP) in aqueous solution." This confirms the removal of an organic pollutant (CIP) from water.

Response B1: Yes
Reasoning B1: The paper focuses solely on electrooxidation/electrocatalytic oxidation for CIP degradation, as evidenced by the abstract: "electrocatalytic degradation of ciprofloxacin (CIP) in aqueous solution." No other coupled technologies (e.g., photocatalysis, PMS activation) are mentioned.

Response B2: No
Anode: Sb-doped SnO2
Reasoning B2: The anode material is Sb-doped SnO2. Antimony (Sb) is not a transition metal, and tin (Sn) is also not a transition metal. Thus, the anode does not contain any transition metal elements.

Response C: Yes
Reasoning C: The study is a laboratory-level experimental work, as indicated by the preparation of electrodes, electrochemical characterization, and optimization of experimental parameters for CIP degradation, as described in the abstract.

Fig. 4. Effectiveness of the 3T+RAG prompting framework in improving article screening performance. a-b Schematic comparison of the two prompt engineering strategies. **a** The Simple prompt provides basic task instructions for screening articles on electrochemical degradation of organic pollutants. **b** The 3T+RAG prompt enhances the standard instructions by incorporating precise domain knowledge (e.g., definitive classification of chemical elements) and requiring a structured, criteria-by-criteria judgement. **c-d** Representative model responses to the simple prompt. Both DeepSeek (**c**) and ChatGPT (**d**) incorrectly return an "Overall Response: Yes" for a study using an Sb-doped SnO_2 anode, erroneously classifying antimony (Sb) or tin (Sn) as transition metals, demonstrating a limitation of the baseline prompt. **e** The corrective response from DeepSeek using the 3T+RAG prompt. By applying the retrieved chemical classification relevant to criterion B2, the model correctly evaluates the anode composition and returns the appropriate screening decision.

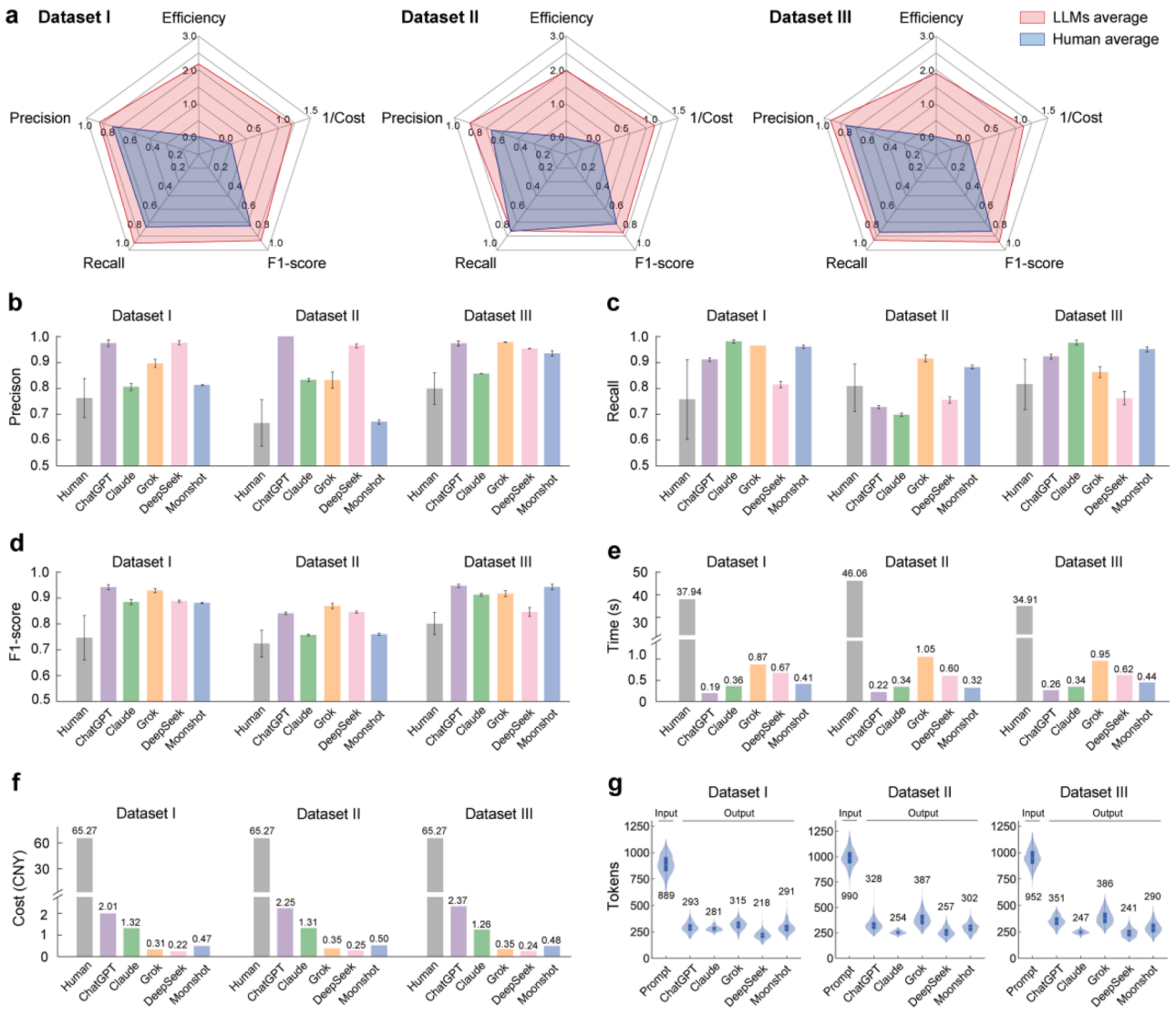


Fig. 5. Quantitative evaluation of the performance, efficiency, and cost of domain-enhanced LLMs and individual human annotators. a Overall results of human annotators (n=9) and domain-enhanced LLMs using the 3T+RAG prompting strategy across three water-related datasets. Human results represent independent screening decisions from individual annotators rather than consensus-based expert adjudication. b-d Detailed performance metrics: precision (b), recall (c), and F1-score (d). Data are presented as mean ± standard deviation (SD) (n=9 for human annotators; n=3 repeated runs per LLM). e Average processing time per article. Time for LLMs was measured using 20 parallel threads. f Average cost for screening 100 articles (unit: Chinese Yuan, CNY). g Token usage distribution for complete prompts and LLM responses. Values indicate mean token counts.

groups across the three datasets, demonstrating that the performance advantage of 3T+RAG remained robust under a more demanding human baseline (Table S6).

Second, the domain-enhanced 3T+RAG LLMs also demonstrated advantages in time efficiency. Under a benchmark setting using 20 parallel threads, they processed articles 79 times faster than human annotators on average (0.50 vs. 39.64 seconds per article; Fig. 5e). The processing times varied across the evaluated LLMs, ranging from 0.22 seconds (ChatGPT-4.1) to 0.96 seconds (Grok-3-mini), reflecting differences in model architecture and application programming interface (API) latency. More importantly, the average processing time of domain-enhanced LLMs decreased exponentially as the number of parallel threads increased (Fig. S2), highlighting a scalability fundamentally unattainable for human processing.

Third, domain-enhanced LLM-based screening showed a substantial cost advantage, with an average cost of 0.91 CNY per 100 articles, corresponding to approximately 1.4% of the estimated human screening cost of 65.27 CNY per 100 articles (Fig. 5f). The cost of individual human screening was estimated using the measured average screening time of 39.64 s per article and the annual compensation levels of the master's, doctoral, and postdoctoral researchers included in the evaluation cohort. Assuming 2,000 working hours per year and equal representation of the three experience levels, the weighted average human screening cost was estimated at 0.65 CNY per article per annotator. On the other hand, the overall cost of LLMs was determined by both token volume and API pricing (Wu et al., 2026). Although input tokens were approximately three times more numerous than output tokens (Fig. 5g), the unit price of output tokens was 4-8 times higher (Table S7), indicating that both input volume and output pricing served as critical cost drivers. Therefore, a balanced optimization strategy that focuses on streamlining the input context while maintaining output efficiency is essential for achieving further cost reductions.

3.4. A multi-agent framework enables decision-level traceability of disagreements

Despite advantages in accuracy, speed, and cost, the single-LLM approach (including general-purpose and domain-enhanced) lacks an

inherent mechanism for identifying uncertain decisions or tracing potential errors, compromising system trustworthiness. To address this challenge, we developed a multi-agent Reviewer-Reviewer-Arbiter (RRA) framework. As illustrated in Fig. 6a, this framework comprises two independent reviewer agents, guided by 3T+RAG prompts to process articles in parallel, and an arbiter agent that performs in-depth analysis of disagreements. Representative examples of reviewer prompts, arbiter prompts, and structured outputs generated within the RRA framework are provided in Texts S11 and S12. The initial classifications are programmatically compared and only inconsistencies trigger arbitration, during which the arbiter examines the full reasoning records to reach a reasoned final decision. This process introduces a taxonomy for error traceability based on the initial results of the reviewer agents. After both reviewers independently evaluate an article, three outcomes are possible: (1) consistent correctness, which is agreement with the ground truth and requires no arbitration; (2) consistent errors, deemed untraceable since the concurrent failure of both agents provides no internal signal for detection; and (3) inconsistent errors, classified as traceable since the disagreement indicates that one agent has made an incorrect judgement, thereby indicating a potential mistake for further arbitration. These three outcome categories can be distinguished only during labelled benchmark evaluation. In real-world deployment, the directly observable quantities are the decisions returned by the two reviewer agents and whether they agree or disagree. In this framework, "traceability" refers specifically to decision-level traceability: the capacity to identify disputed decisions through reviewer outputs and reasoning, rather than label-free verification of correctness.

Evaluation on three datasets demonstrated that consistent correctness constituted the majority of outcomes, with both models reaching the correct decision for approximately 89.2% of the articles, supporting the reliable application of the proposed framework. Traceable errors accounted for 3.2% to 12.0% across the three datasets, while untraceable errors remained minimal, ranging from 1.1% to 3.6% (Fig. 6b). The linguistic habits of the reviewer agents, as reflected in their high-frequency word usage (Tables S8-S10), further confirm their domain-specific focus. To visualize these linguistic patterns and compare them across all evaluated LLMs, we generated word clouds from their responses for each dataset (Figs. S3-S5). These visualizations not only corroborate the high-frequency word analysis but also intuitively reveal that, despite architectural differences, all models converged on core domain terminology. These results support the use of reviewer disagreement as a practical warning signal under the evaluated conditions, while untraceable error (consistent error) cannot be measured during deployment without external labels. Analysis of the sources of

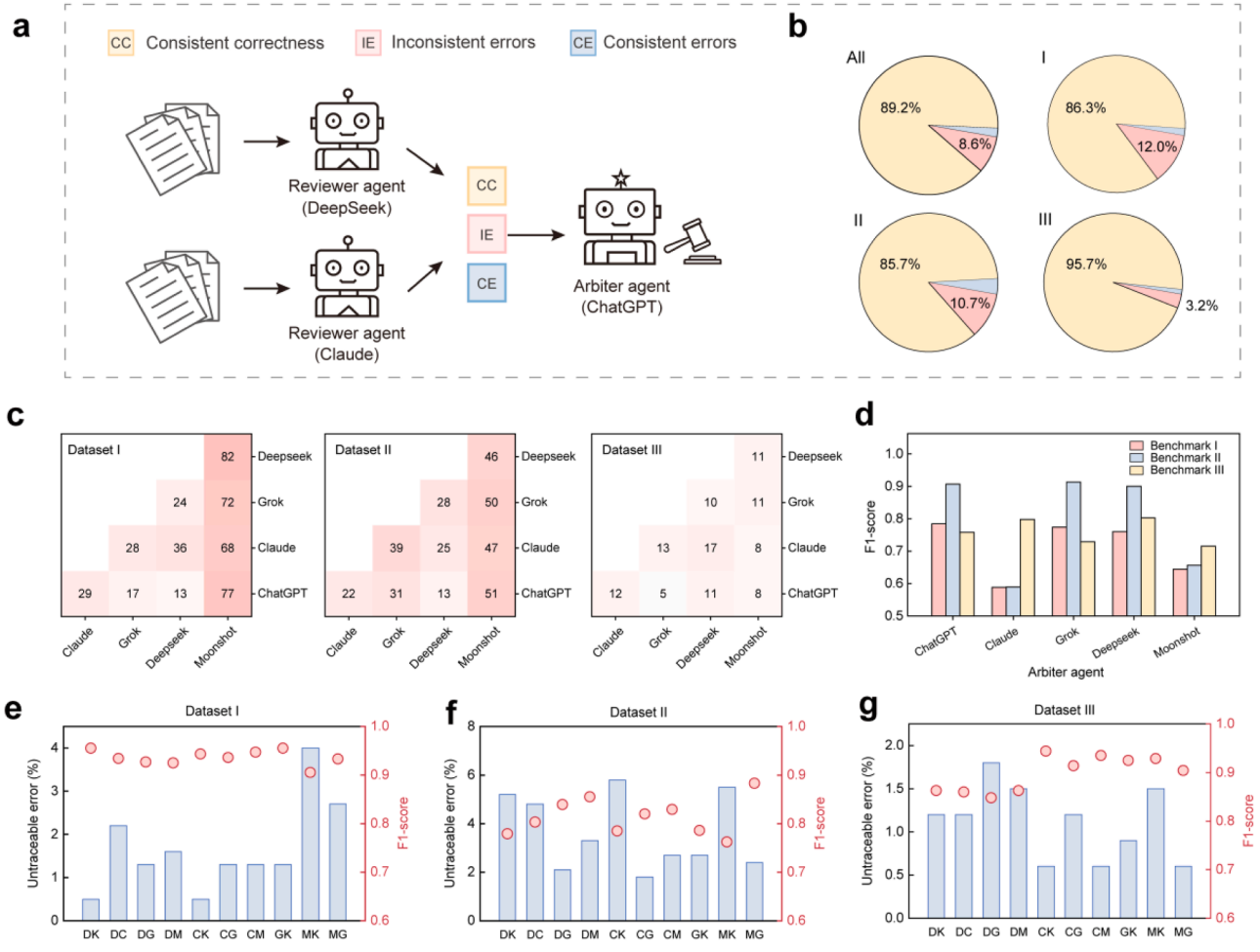


Fig. 6. Performance and error analysis of the Reviewer-Reviewer-Arbiter (RRA) multi-agent framework. **a** Schematic of the RRA decision workflow. Input articles are evaluated by two different reviewer agents, resulting in three possible outcomes: consistent correctness (yellow), where both reviewers agree and are correct; inconsistent errors (pink), where reviewers disagree; and consistent errors (blue), where both reviewers agree but are incorrect. Cases of inconsistent errors are resolved by an arbiter agent. **b** Proportional distribution of reviewer outcomes. Pie charts show the proportions of consistent correctness, inconsistent errors, and consistent errors across all datasets combined (All) and for each individual dataset (I, II, III). **c** Number of inconsistent errors by different combinations of reviewer agents. **d** Arbiter performance on disagreement benchmark datasets. **e-g** RRA framework performance for Dataset I (**e**), Dataset II (**f**), and Dataset III (**g**). Red lines indicate overall F1-score of RRA framework, using ChatGPT as the arbiter; blue bars show the untraceable error rate for each reviewer combination. Reviewer agent abbreviations: D (DeepSeek-V3), C (ChatGPT-4.1), G (Grok-3-mini), M (Moonshot-v1-8k), and K (Claude-3.5-haiku).

traceable errors reveals that the distribution of disagreements is model-dependent. For Datasets I and II, reviewer pairs including Moonshot-v1-8k accounted for most disagreements (Fig. 6c), which could be attributed to the lower individual performance of Moonshot (Fig. 5d). In contrast, for Dataset III, the DeepSeek-V3-based reviewer accounted for the most disputes (Fig. 6c).

To identify the best arbiter model, we aggregated these disputed articles into three disagreement benchmarks and evaluated the performance of five prevalent LLMs as arbiter agents. The results showed that ChatGPT-4.1, Grok-3-mini, and DeepSeek-V3 demonstrated superior arbitration performance across all three test sets compared to Claude-3.5-haiku and Moonshot-v1-8k (Fig. 6d). Based on the evaluation performance and processing speed, ChatGPT-4.1 was selected as the arbiter for all subsequent investigations (Figs. 5e and 6d). Notably, the observed performance differences also reflect disparities in model scale in addition to vendor-specific architectural limitations. For instance, Claude-3.5-haiku is a lightweight variant optimized for latency and cost-efficiency, whereas ChatGPT-4.1 represents a more capable model. Similarly, Moonshot-v1-8k is a compact model compared with the 671-billion-parameter DeepSeek-V3. In addition, the arbitration mechanism could not correct all errors due to the limited capacities of current LLMs and the inherent ambiguity of the task. Therefore, to achieve higher performance, we recommend incorporating human oversight, with researchers focusing primarily on disputed articles while also performing targeted quality control on agreed decisions (Fig. S6) (Wang et al., 2025; Wu et al., 2026).

The principal goal of the RRA framework was to achieve both high screening performance and a low untraceable error ratio, thereby enhancing the overall traceability and reliability of the screening process. Having identified ChatGPT-4.1 as the arbiter, we next investigated the impact of different reviewer-agent combinations to determine the optimal configuration of the RRA framework. On Dataset I, most reviewer combinations achieved high F1-scores (>0.90), with the DeepSeek-V3@Claude-3.5-haiku (DK) and ChatGPT-4.1@Claude-3.5-haiku (CK) reviewer configurations exhibiting the lowest untraceable error rates, making them the recommended options on this dataset (Fig. 6e). On Dataset II, F1-scores varied significantly across reviewer combinations. Among these, the Moonshot-v1-8k@Grok-3-mini (MG) configuration achieved the highest F1-score (0.883) while maintaining a relatively low untraceable error rate (2.4%), rendering it a favorable option for this dataset (Fig. 6f). In the case of Dataset III, the proportion of untraceable errors was the lowest among all three datasets. Following the same selection principle, the ChatGPT-4.1@Claude-3.5-haiku (CK) combination was recommended due to its high F1-score (0.944) and minimal untraceable errors (0.6%) (Fig. 6g). Based on the optimal reviewer and arbiter models identified above, the RRA framework can achieve optimal classification performance while reducing traceable errors, thereby enhancing the traceability and trustworthiness of model decisions.

4. Conclusion

This study demonstrates that domain-enhanced strategies enable leverage general-purpose LLMs to perform literature screening in water research with high accuracy and efficiency. By integrating the structured 3T prompt, retrieval-augmented generation, and multi-agent review and arbitration, the framework enables general-purpose LLMs to assess multidimensional, topic-specific eligibility criteria with improved accuracy, efficiency, and reproducibility. The 3T+RAG strategy provides explicit screening pathways and domain-specific knowledge, while allowing incorrect classifications to be traced to specific eligibility criteria. Benchmark evaluations across multiple LLMs, together with comparisons against individual human annotation and three-annotator majority-vote consensus, provided complementary reference points for assessing both model-level screening performance and agreement with human judgement. The results showed that the proposed strategy improved screening accuracy while maintaining rapid processing and low operational costs. These findings demonstrate that semantic understanding and domain constraints can complement conventional database retrieval by improving the accuracy of literature screening.

The multi-agent RRA framework provides an additional quality-control layer by recording reviewer decisions, identifying disagreement cases, and distinguishing between traceable and untraceable errors. This error taxonomy supports a more targeted form of human oversight. Rather than manually screening every retrieved article, researchers can concentrate on resolving disputed cases and auditing consensus classifications. The proposed framework therefore complements, rather than replaces, conventional database retrieval by serving as a qualification layer between broad literature searches and downstream scientific analysis. It could facilitate the construction of more consistent, scalable, and auditable literature datasets.

The present study was limited to eligibility screening in three water-treatment domains and did not address structured data extraction, evidence synthesis, mechanistic interpretation, pattern discovery, or hypothesis generation. In addition, the RAG repositories were specifically developed for the investigated topics. Future work should therefore evaluate the transferability of the framework to broader environmental domains, develop more efficient strategies for repository construction and updating, and extend the workflow to downstream knowledge extraction and scientific discovery tasks.

CRediT authorship contribution statement

Runzhi Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Conceptualization. **Xinlu Lin:** Data curation. **Baoyi Zhong:** Methodology, Data curation. **Jiaxuan Li:** Methodology. **Ruixing Huang:** Writing – review & editing. **Yu Huang:** Methodology. **Yajuan Cai:** Writing – review & editing. **Chotiwat Jantarakasem:** Formal analysis. **Yang Xiang:** Supervision. **Konstantinos Plakas:** Writing – review & editing. **Jun Ma:** Supervision, Conceptualization. **Yumeng Zhao:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2026.126813.

Data availability

Data will be made available on request.

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